

# Counting cones over reducible cubic scrolls

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**Abstract.** Here we construct an explicit parameter space for cones in  $\mathbb{P}^n$  for  $4 \leq n \leq 6$  with vertices of codimension 4 and 5, respectively (when this makes sense), over a reducible cubic scroll of codimension 2 in a hyperplane (in  $\mathbb{P}^n$ ) and use it to compute the numbers of them that are incident to the appropriate number of linear spaces.

**Keywords:** reducible cubic scroll, Hilbert scheme, cones, compactification.

**Mathematical subject classification:** Primary: 14N10, 14N05.

## 1 Introduction

In enumerative geometry, one of the central problems since the time of Chasles, Zeuthen and Schubert is to determine the number of lines, curves, or more generally algebraic-geometric objects satisfying certain geometric conditions, e.g. the number of rational plane curves of a given degree  $e$  passing through  $3e - 1$  generic points in the plane. The usual strategy to solve enumerative questions is to find a suitable parameter space and express the number sought as the degree of a class in the Chow ring of the involved parameter space. One way to get rid of the difficulty of find the Chow ring and solve the given enumerative question is provided by the Bott's residual formula, which was used by *Vainsencher-Xavier* in [7] to determine the numbers for reducible cubic scrolls incident to (the appropriate number) of linear spaces and was then also used by *Levcovitz-Vainsencher-Xavier* in [2] to enumerate cones over cubic scrolls. In this work we will use the Bott's residual formula to compute cones in  $\mathbb{P}^n$  for  $4 \leq n \leq 6$  over a reducible cubic scrolls of codimension 2 in a hyperplane (in  $\mathbb{P}^n$ ) incident to the appropriate number of linear spaces. The difference between

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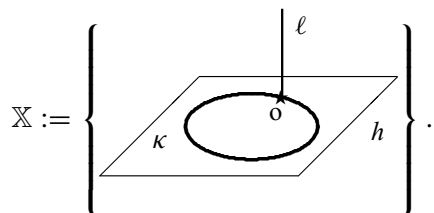
our work and that in [2] lies in the fact that the base of the cones that we study here are *reducible* cubic scrolls. Moreover, the authors in [7] only compute numbers for reducible cubic scrolls and, they made no mention of how to compute numbers for cones over it.

In order to expose the aim of our work in more detail let us first introduce some notations. Let  $\mathbb{H}$  be the component of the Hilbert scheme whose generic point is a cubic rational normal scroll of codimension 2 in  $\mathbb{P}^n$ . Let  $\mathbb{H}_{redb}$  be the subscheme of  $\mathbb{H}$  parametrizing the reducible ones. As noted in [7], a general reducible cubic scroll of codimension 2 in  $\mathbb{P}^N$  is the union of an  $(N - 2)$ -plane  $\ell$  with an  $(N - 2)$ -dimensional quadric  $\kappa$  meeting along an  $(N - 3)$ -plane  $o$ , just as in the case of twisted cubics for  $N = 3$ . The authors also observed (see closing remark at p. 650 in [7]) that for  $N > 5$  all rational normal cubic scrolls of codimension 2 in  $\mathbb{P}^N$  are cones.

Now let  $\mathbb{G}[r, N]$  be the Grassmann variety of  $r$ -planes in  $\mathbb{P}^N$ . Set  $x = (o, \ell, \kappa \subset h)$  and consider

$$\mathbb{X} = \{x \mid o \subset \ell \cap \kappa, \ell \in \mathbb{G}[N - 2, N], o \in \mathbb{G}[N - 3, N] \\ \text{and } \kappa \text{ is a quadric in } h \in \mathbb{G}[N - 1, N]\}.$$

Pictorially for  $N = 3$ .



We have the rational map,

$$\sigma : \mathbb{X} \cdots \longrightarrow \mathbb{H}_{redb} \subset \mathbb{H}, \quad (o, \ell, \kappa) \longmapsto \ell \cup \kappa. \quad (1)$$

It fails to be regular along the subvariety  $\mathbb{Y}$  of  $\mathbb{X}$  described in (8). Let  $\mathbb{X}'$  be the blowup of  $\mathbb{X}$  along  $\mathbb{Y}$ .

In this work we construct a flat family of closed subschemes over  $\mathbb{X}'$  extending what we get naturally by base change to  $\mathbb{X} \setminus \mathbb{Y}$  from the universal family on  $\mathbb{H}$ . Or equivalently (follows from the universal property on  $\mathbb{H}$ ), we get a morphism  $\mathbb{X}' \longrightarrow \mathbb{H}_{redb}$  extending  $\sigma$  (in accordance with remark 2.2 in [7]). Finally (cf. Section 4) we use the parameter space  $\mathbb{X}'$  to compute the number of cones in  $\mathbb{P}^n$  for  $4 \leq n \leq 6$  with vertices of dimension  $v = 0, 1, 2$ , respectively (when this makes sense), over a reducible cubic scroll of codimension 2 in a hyperplane (in  $\mathbb{P}^n$ ) incident to the appropriate number of linear spaces.

## 2 Parameter space for cones over reducible cubic scrolls

### 2.1 The construction of $\mathbb{X}$

Denote by  $\mathbb{G}[d, n]$  the grassmannian of  $d$ -linear subspaces of  $\mathbb{P}^n$ . Let  $\mathbf{B}$  be a subvariety of dimension  $b$  in  $\mathbb{P}^n$  which spans  $\langle \mathbf{B} \rangle \in \mathbb{G}[b+2, n]$  and  $\mathbf{V} \in \mathbb{G}[v, n]$  such that  $\mathbf{V} \subset \mathbb{P}^n \setminus \langle \mathbf{B} \rangle$  and  $v = n - b - 3$ . Then the *cone* over  $\mathbf{B}$  with vertex  $\mathbf{V}$  is the union of the  $\Lambda \in \mathbb{G}[v+1, n]$  such that  $\Lambda = \langle \mathbf{V}, p \rangle$  with  $p \in \mathbf{B}$  (see p. 32 in [3]). Next we introduce some bundles and varieties in order to describe a parameter space for cones in  $\mathbb{P}^n$  ( $n = v + b + 3$ ) with vertex in  $\mathbb{G}[v, n]$ .

Let  $\mathcal{F}$  denote the space of linear forms in the variables  $\underline{x} = x_1, \dots, x_{n+1}$ . And let

$$0 \longrightarrow \mathcal{T} \longrightarrow \mathcal{F} \longrightarrow \mathcal{V} \longrightarrow 0$$

be the dual of the tautological sequence over the grassmannian of vertices  $\mathbb{G}[v, n]$ , where the fiber of  $\mathcal{T}$  over  $\mathbf{V} \in \mathbb{G}[v, n]$  is given by the space of linear forms that vanish on the  $v$ -plane  $\mathbf{V}$ . So the rank of  $\mathcal{T}$  is  $n - v$ .

Let  $\mathbb{G}(2, \mathcal{T}) \longrightarrow \mathbb{G}[v, n]$  be the Grassmann bundle of rank-2 vector subspaces of the fibers of  $\mathcal{T} \longrightarrow \mathbb{G}[v, n]$  with tautological sequence

$$0 \longrightarrow \mathcal{L}_{\mathcal{T}} \longrightarrow \mathcal{T} \longrightarrow \overline{\mathcal{T}} \longrightarrow 0. \quad (2)$$

Let  $S_d \mathcal{T}$  denotes the  $d$ -symmetric power of  $\mathcal{T}$ . Let  $S_2 \mathcal{T}^\perp$  denote the vector bundle of rank  $2(n - v) - 1$  over  $\mathbb{G}(2, \mathcal{T})$  whose fiber over  $\ell$ ,  $S_2 \mathcal{T}^\ell$ , is equal to the subspace of  $S_2 \mathcal{T}$  of forms of degree 2 that vanish on the codimension 2 linear subspace  $\ell$  (by duality we identify  $\ell \in \mathbb{G}(2, \mathcal{T})$  with the codimension 2 linear subspace also denoted by  $\ell \subset \mathbb{P}(\mathcal{T}^\vee)$ ), ie.

$$S_2 \mathcal{T}^\perp := \text{Ker}(S_2 \mathcal{T} \rightarrow S_2 \overline{\mathcal{T}}).$$

Let

$$0 \longrightarrow \mathcal{O}_{\mathcal{T}}(-1) \longrightarrow \mathcal{T} \longrightarrow \mathcal{H} \longrightarrow 0 \quad (3)$$

be the tautological sequence over  $\mathbb{P}(\mathcal{T})$ , where the fiber of  $\mathcal{O}_{\mathcal{T}}(-1)$  over  $h \in \mathbb{P}(\mathcal{T})$  is given by the space of linear forms that vanish on the hyperplane  $h$ . So the rank of  $\mathcal{O}_{\mathcal{T}}(-1)$  is 1.

We still denote by  $S_2 \mathcal{T}^\perp$ ,  $\mathcal{L}_{\mathcal{T}}$  and  $\mathcal{O}_{\mathcal{T}}(-1)$  the pullback of the previous bundles to  $\mathbb{G}(2, \mathcal{T}) \times \mathbb{P}(\mathcal{T})$  via the respective projection maps.

Consider the vector bundle

$$\mathcal{E} := \frac{S_2 \mathcal{T}^\perp}{\mathcal{O}_{\mathcal{T}}(-1) \otimes \mathcal{L}_{\mathcal{T}}} \quad (4)$$

over  $\mathbb{G}(2, \mathcal{T}) \times \mathbb{P}(\mathcal{T})$ .

Note that the fiber of  $\mathcal{E}$  over  $(\ell, h)$  is the quotient vector space  $S_2\mathcal{T}^\ell/h \cdot \ell$  where  $h \cdot \ell$  denote the 2-dimensional subspace of quadrics  $\langle hh_1, hh_2 \rangle$ , here  $h_1$  and  $h_2$  are linear forms determining the codimension 2 linear subspace  $\ell$ . Also note that when the hyperplane  $h$  does not contain the distinguished codimension 2 linear subspace  $\ell$ , we have a well defined  $(n - v - 4)$ -plane given by the intersection of the hyperplane  $h \in \mathbb{P}(\mathcal{T})$  with the codimension 2 linear subspace  $\ell$ . Nevertheless, when the hyperplane “swallows” the codimension 2 linear subspace  $\ell$ , we miss the  $(n - v - 4)$ -plane. In order, to get a well defined  $(n - v - 4)$ -plane  $\mathfrak{o} \subset \ell \subset h$ , we consider the variety  $\mathbb{B}$  given by the blowup of  $\mathbb{G}(2, \mathcal{T}) \times \mathbb{P}(\mathcal{T})$  along the incidence subvariety

$$\Sigma = \{(\ell, h) \in \mathbb{G}(2, \mathcal{T}) \times \mathbb{P}(\mathcal{T}) \mid \ell \subset h\}. \quad (5)$$

Let  $\mathcal{E}$  also denote the pullback of the vector bundle (4) to  $\mathbb{B}$  via the blow-up map  $\mathbb{B} \rightarrow \mathbb{G}(2, \mathcal{T}) \times \mathbb{P}(\mathcal{T})$  and let  $\mathbb{X} = \mathbb{P}(\mathcal{E})$ .

Now and henceforth for the sake of concreteness we will describe the construction of the parameter space for cones in  $\mathbb{P}^n$  ( $n = 4, 5, 6$ ) with vertices of dimension  $v = 0, 1, 2$ , respectively. Thus the set of vertices has codimension 4 in each case.

**Remarks 2.1.** Fix the vertex  $\mathbf{V} \subset \mathbb{P}^n$  defined by the linear forms  $x_1, \dots, x_4$ , so  $\mathcal{T}_{\mathbf{V}} = [x_1, \dots, x_4]$ . Put  $\mathcal{G} = \mathcal{T}_{\mathbf{V}}$ .

- Since  $\Sigma_{\mathbf{V}}$  is normally flat over  $\mathbb{G}(2, \mathcal{G})$ , the fiber of the blow-up is isomorphic to the blow-up of the fiber. So, for simplicity, we explain the procedure for the fiber  $\mathbb{X}_{\ell}$  over the distinguished line  $\ell = \langle x_1, x_2 \rangle$ .
- $\Sigma$  is a projective  $\mathbb{P}^1$ -bundle over  $\mathbb{G}(2, \mathcal{T})$ . In fact the fiber of  $\Sigma$  over  $(\mathbf{V}; \ell)$  is given by  $\mathbb{P}(\mathcal{G}^\ell)$  where  $\mathcal{G}^\ell$  is the space of linear forms in  $\mathcal{G}$  that vanished over  $\ell$ .
- Let  $\widetilde{\mathbb{P}(\mathcal{G})}$  be the blow-up of  $\mathbb{P}(\mathcal{G})$  along the pencil of planes through the distinguished line  $\ell = \langle x_1, x_2 \rangle$ , i. e. the closure of the graph of the rational map

$$\begin{aligned} \psi: \mathbb{P}(\mathcal{G}) &\cdots \longrightarrow \ell \cong \mathbb{P}^1 \subset \mathbb{P}^3 = \mathbb{P}(\mathcal{G}^\vee) \\ h &\longmapsto h \cap \ell. \end{aligned}$$

Then the fiber of the exceptional divisor  $\mathbb{E}$  over  $h$  is given by  $\mathbb{E}_h = \mathbb{P}(\mathcal{G}/\langle x_1, x_2 \rangle)$  and corresponds to all possible choices of a point “o” in

the line  $\ell \subset h$ .

$$\mathbb{E}_h := \left\{ \begin{array}{c} \text{Diagram: A parallelogram representing a plane } h. \text{ Inside, a horizontal line segment represents } \ell. \text{ A point } o \text{ is marked on } \ell. \end{array} \right\}.$$

- Having in mind that the Hilbert polynomial of twisted cubics in  $\mathbb{P}^3 = \mathbb{P}(\mathcal{G}^\vee)$  is  $3t - 1$  we can expect (in the general case) that the ideal defining such a curve will be generated in degree 2, 3 by 3 quadratics forms, 10 cubic forms, respectively.

On the other hand a point  $x \in \mathbb{X}$  (lying over the vertex  $\mathbf{V}$ ) correspond to the following data  $x = (\ell, h; o; \bar{q})$  where  $o \in \ell \cap h$ ,  $(\ell, h) \in \mathbb{G}(2, \mathcal{G}) \times \mathbb{P}(\mathcal{G})$  and  $\bar{q} \in \mathbb{P}(S_2\mathcal{G}^\ell/h \cdot \ell)$ . In the following (as in the rest of the text) we denoted by the same letter  $\mathbb{X}$  the fiber of  $\mathbb{X}$  over  $\mathbf{V}$ .

The map described in the following tell us that we can get the appropriate number of quadrics from any point in  $\mathbb{X}$ . In fact, consider the following well defined regular map

$$\begin{aligned} \alpha : \mathbb{X} &\longrightarrow \mathbb{G}(3, S_2\mathcal{G}) \\ (b, \bar{q}_3) &\longmapsto \langle h \cdot h_1, h \cdot h_2, q_3 \rangle, \end{aligned} \quad (6)$$

with  $b$  lying over  $(\ell = \langle h_1, h_2 \rangle, h) \in \mathbb{G}(2, \mathcal{G}) \times \mathbb{P}(\mathcal{G})$  and  $\bar{q}_3 \in \mathbb{P}(S_2\mathcal{G}^\ell/h \cdot \ell)$ .

On the other hand, the rational map  $\beta$  given by (notations as above).

$$\begin{aligned} \beta : \mathbb{X} &\cdots \longrightarrow \mathbb{G}(10, S_3\mathcal{G}) \\ (b, \bar{q}_3) &\longmapsto hh_1 \cdot \mathcal{G} + hh_2 \cdot \mathcal{G} + q_3 \cdot \mathcal{G}. \end{aligned} \quad (7)$$

It's not defined over the subvariety  $\mathbb{Y}$  of  $\mathbb{X}$  described in (8).

From now on, unless otherwise specified, for simplicity, we will omit from the notation of points in  $\mathbb{X}$ , the fixed vertex  $\mathbf{V}$  defined by the linear forms  $x_1, \dots, x_4 \in \mathcal{T}_{\mathbf{V}} = \mathcal{G}$ .

## 2.2 Closed orbits in $\mathbb{X}$

The computation of closed orbits for  $\mathbb{X}$  under the action of the group  $\mathbb{PGL}(3)$  is one of the basic ingredients that we will use in the proof of Lemmas 2.6, 2.9 and Proposition 2.8.

**Lemma 2.2.** *There are exactly two closed orbits in  $\mathbb{X}$ , represented by the points*

$$\begin{aligned} o_1 &= (\langle x_1, x_2 \rangle, x_1; \langle x_1, x_2, x_3 \rangle; \overline{x_1 x_3}) \quad \text{and} \\ o_2 &= (\langle x_1, x_2 \rangle, x_1, ; \langle x_1, x_2, x_3 \rangle; \overline{x_2^2}). \end{aligned}$$

**Proof.** We may start by setting

$$o = \langle x_1, x_2, x_3 \rangle \in \ell = \langle x_1, x_2 \rangle \subset h = x_1.$$

Note that the stabilizer of the point determined by  $(\ell, h; o)$  in  $\mathbb{E}_h$  is given by

$$\begin{pmatrix} a_{1,1} & 0 & 0 & 0 \\ a_{2,1} & a_{2,2} & 0 & 0 \\ a_{3,1} & a_{3,2} & a_{3,3} & 0 \\ a_{4,1} & a_{4,2} & a_{4,3} & a_{4,4} \end{pmatrix}$$

where  $a_{i,i} \neq 0$  for  $1 \leq i \leq 4$ . Since the stabilizer of  $(\ell, h; o) \in \mathbb{E}_h$  is a solvable group, the representatives of closed orbits in  $\mathbb{X}$  correspond to fixed points in  $\mathbb{X}_{(\ell, h; o)}$  under the action of the stabilizer of  $(\ell, h; o) \in \mathbb{E}_h$  (see p. 340 in [5] and [6]).

The general element  $\bar{q} \in \mathbb{X}_{(\ell, h; o)}$  has a representative in the form

$$q = c_1 x_1 x_3 + c_2 x_1 x_4 + c_3 x_2^2 + c_4 x_2 x_3 + c_5 x_2 x_4, \quad [c_1, c_2, c_3, c_4, c_5] \in \mathbb{P}^4.$$

If  $c_3 \neq 0$  we may act with  $x_3 \mapsto tx_3, x_4 \mapsto tx_4$  and get  $o_2$  in the orbit's closure. If  $c_3 = 0$  and  $c_4 \neq 0$  or  $c_5 \neq 0$  we make the change of coordinates,  $x_i \mapsto x_i + x_2$ , ( $i = 3$  or  $4$ ) and reduce to the previous case. Finally, if  $q = c_1 x_1 x_3 + c_2 x_1 x_4$  we may act with  $x_4 \mapsto tx_4$  and show that  $q = c_1 x_1 x_3$ . This produces  $o_1$ .  $\square$

**Remark 2.3.** The orbit  $o_2$  it is not used in the construction of our compactification. Indeed, the multiplication map studied in (9) below is of maximal rank in  $o_2$  and the rational map  $\sigma : \mathbb{X} \cdots \longrightarrow \mathbb{H}_{redb} \subset \mathbb{H}$  (cf.(1)) is regular there. But it will appear in the computation of Bott's localization formula in the last section.

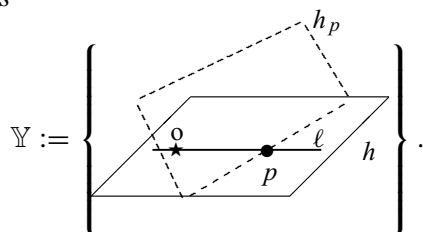
### 2.3 Description of $\mathbb{Y}$

Notation as in (2) and Remarks 2.1. Let  $\mathbb{E} \subset \mathbb{B}$  be the exceptional divisor (lying over a fixed vertex  $\mathbf{V}$ ). Then we may think of a point  $b \in \mathbb{E}$  as a triple  $(\ell, h; o)$ , where  $o$  is a point in the line  $\ell \subset h \in \check{\mathbb{P}}^3 = \mathbb{P}(\mathcal{G})$ . Let  $\mathcal{L}$  be the tautological bundle of  $\mathbb{G}(2, \mathcal{G})$ .

**Lemma 2.4.** *Let  $\mathbb{Y}$  be subvariety of  $\mathbb{X}$  laying over  $\mathbb{E}$  given by*

$$\mathbb{Y} = \{(\mathbf{b}, \bar{q}) \in \mathbb{X}_{\mathbf{b}} \subset \mathbb{X} \mid \mathbf{b} = (\ell, h; \mathbf{o}) \in \mathbb{E}, q = hh_p, \bar{h}_p \in \mathbb{P}(\mathcal{G}/\mathcal{L}_{\ell})\}, \quad (8)$$

*which is illustrated as*



*Then we have the following.*

1. *Let  $\Gamma$  be the incidence subvariety given by  $\{(p, \ell) \in \mathbb{P}^3 \times \mathbb{G}(2, \mathcal{G}) \mid p \in \ell\}$ . Then there is a closed embedding*

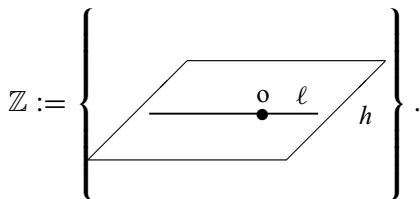
$$\begin{aligned} i : \mathbb{E} \times_{\mathbb{G}(2, \mathcal{G})} \Gamma &\longrightarrow \mathbb{Y} \subset \mathbb{X} \\ ((\ell, h; \mathbf{o}), (p, \ell)) &\longmapsto ((\ell, h; \mathbf{o}), \overline{hh_p}) \end{aligned}$$

*where  $h_p$  is a linear form determining the point  $p \in \ell$ .*

2.  *$\mathbb{Y}$  is a  $\mathbb{P}^1$ -bundle over  $\mathbb{E}$  with fiber over  $(\ell, h; \mathbf{o})$  given by  $\mathbb{P}(\mathcal{G}/\mathcal{L}_{\ell})$ .*
3. *The subvariety  $\mathbb{Y}$  is  $\mathrm{PGL}(3)$ -invariant.*
4. *The orbit of  $((x_1, x_2), x_1, \overline{x_1 x_3})$  is the unique closed orbit in  $\mathbb{Y}$ . In fact, this closed orbit is isomorphic to the variety of complete flags*

$$\mathbb{Z} = \{(\mathbf{o}, \ell, h) \in \mathbb{P}^3 \times \mathbb{G}(2, \mathcal{G}) \times \check{\mathbb{P}}^3 \mid \mathbf{o} \in \ell \subset h\}.$$

*which is illustrated as*



In Lemma 2.6 below we will give a scheme structure for  $\mathbb{Y}$  (i.e. equations defining  $\mathbb{Y}$  locally). We note that the key point in it follows from the following lemma.

**Lemma 2.5.** *Let  $\mathbb{V}$  be a smooth projective variety.  $\mathbb{G}$  a connected linear algebraic group acting on  $\mathbb{V}$  and  $\mathbb{W}$  a closed irreducible smooth  $\mathbb{G}$ -invariant subvariety of  $\mathbb{V}$ . Let  $\mathbb{W}'$  be a closed,  $\mathbb{G}$ -invariant subscheme of  $\mathbb{V}$  such that  $\mathbb{W} \subseteq \mathbb{W}'$ . Suppose that for each closed orbit of  $\mathbb{W}'$  there is a representative  $\mathfrak{o} \in \mathbb{W}$  such that the dimension of the tangent space of  $\mathbb{W}'$  at  $\mathfrak{o}$  is equal to  $\dim \mathbb{W}$ . Then  $\mathbb{W} = \mathbb{W}'$ .*

**Proof.** See p. 341 in [5]. □

**Lemma 2.6.** *Let  $\mathcal{A}$  be the pullback of the rank three tautological bundle of  $\mathbb{G}(3, S_2\mathbb{G})$  via  $\alpha$  as in (6) and let*

$$\mathcal{A} \otimes \mathbb{G} \xrightarrow{\mu} S_3\mathbb{G} \quad (9)$$

*be induced by multiplication. Then the generic rank of  $\mu$  is ten and the scheme of zeros of  $\bigwedge^{10} \mu$  is equal to the subvariety  $\mathbb{Y}$  described in Lemma 2.4.*

**Proof.** Put  $\mathbb{Y}' = Z(\bigwedge^{10} \mu)$ . One checks at once that  $\mathbb{Y} \subseteq \mathbb{Y}'$  holds. Let  $\mathfrak{o}_1 = (\langle x_1, x_2 \rangle, x_1; \overline{x_1 x_3})$  be a representative of the closed orbit  $\mathbb{Z} \subset \mathbb{Y}$ . We will show the equality of the tangent spaces  $T_{\mathfrak{o}_1} \mathbb{Y} = T_{\mathfrak{o}_1} \mathbb{Y}'$ . By lemma 2.5, this implies that  $\mathbb{Y}$  is the unique component of  $\mathbb{Y}'$  through  $\mathfrak{o}_1$ . Since  $\mathbb{Y}'$  is  $\mathrm{PGL}(3)$ -invariant, any component must contain the unique closed orbit, whence  $\mathbb{Y} = \mathbb{Y}'$  as asserted. Note that it will suffice to see that  $\dim T_{\mathfrak{o}_1} \mathbb{Y}' \leq \dim T_{\mathfrak{o}_1} \mathbb{Y}$  holds. Hence we need only to show that the ideal of  $\mathbb{Y}'$  in a neighborhood of the point  $\mathfrak{o}_1$  contains  $\mathrm{codim} \mathbb{Y} = 4$  elements with independent linear terms.

For this we consider a local matrix representation of  $\mu$ . We pick an appropriate coordinate chart for the fiber  $\mathbb{X}_0$  of  $\mathbb{X}$  over  $\ell \in \mathbb{G}(2, \mathbb{G})$  where

$$\ell =: x_1 = x_2 = 0.$$

Let  $U$  be the standard affine neighborhood of  $\mathfrak{o}_1$  in  $\mathbb{X}_0$  with coordinate function  $a_i, i = 1, \dots, 4$  and  $b_j, j = 1, \dots, 3$ . The restriction of the bundle  $\mathcal{A}$  to  $U$  is trivial, generated by

$$\begin{cases} q_1 = (x_1 + b_1 x_2 + b_2 b_3 x_3 + b_3 x_4) x_1, \\ q_2 = (x_1 + b_1 x_2 + b_2 b_3 x_3 + b_3 x_4) x_2, \\ q_3 = x_1 x_3 + a_1 x_1 x_4 + a_2 x_2^2 + a_3 x_2 x_3 + a_4 x_2 x_4 \end{cases} \quad (10)$$

where

$$q_1 = h \cdot x_1 \quad q_2 = h \cdot x_2.$$



The local representation of the map  $\mu$  may be computed as a  $12 \times 20$  matrix in the form

$$M = \begin{pmatrix} I_9 & \star \\ 0 & R \\ 0 & 0 \end{pmatrix} \quad (11)$$

where  $I_9$  denotes an identity block of size 9 and  $R$  is a row matrix that span the Fitting ideal  $\mathcal{J}$ , of  $10 \times 10$  minors of  $\mu$ . So the locus where  $\mu$  drops rank is smooth and is given locally by the ideal

$$\mathcal{J} = \langle a_2, a_3 - b_1, a_4 - a_1 b_1, b_3 \rangle.$$

This shows that  $T_{o_1} \mathbb{Y}'$  is dimension at most  $11 - 4 = 7 = \dim \mathbb{Y}$ . □

**Remark 2.7.** Solving the equations that define the ideal  $\mathcal{J}$ , we have the following four relations that determine  $\mathbb{Y}$  locally,

$$a_2 = 0, a_3 = b_1, a_4 = a_1 b_1, b_3 = 0. \quad (12)$$

Substituting these relations in (10), we can see that the quadrics acquire a common component, namely the plane given by

$$x_1 + b_1 x_2.$$

Note that, this plane contains the distinguished line  $x_1 = x_2 = 0$ . The interesting part of this “net” of planes is  $\langle x_1, x_2, x_3 + a_1 x_4 \rangle$ . It defines the new point  $p = [0 : 0 : -a_1 : 1] \in \ell$ .

In fact, the fiber  $\mathbb{Y}_\ell$  is isomorphic to the variety

$$\mathbb{Y}_\ell = \{ (o, p, h) \in (\mathbb{P}^3)^{\times 2} \times \check{\mathbb{P}}^3 \mid o, p \in \ell \subset h \}$$

and we can see that  $\mathbb{Y}_\ell$  is a tower of  $\mathbb{P}^1$  fiber bundles.

$$\mathbb{Y}_\ell := \left\{ \begin{array}{c} \text{Diagram of a plane } h \text{ containing a line } \ell. \text{ The line } \ell \text{ contains points } o \text{ (marked with a star) and } p \text{ (marked with a dot).} \end{array} \right\}.$$

## 2.4 Description of $\mathbb{X}'$

Notation as in Remarks 2.1.

**Proposition 2.8.** *Let  $\beta : \mathbb{X} \cdots \longrightarrow \mathbb{G}(10, S_3\mathcal{G})$  be the rational map defined in (7). Then the blowup  $\mathbb{X}'$  of  $\mathbb{X}$  along  $\mathbb{Y}$  embeds in  $\mathbb{X} \times \mathbb{G}(10, S_3\mathcal{G})$  as the closure of the graph of this rational map.*

**Proof.** Let  $\mathbb{X}' \rightarrow \mathbb{X}$  be the blowup map of  $\mathbb{X}$  along  $\mathbb{Y}$ . Since flatness is an open condition, it suffices to restrict to an affine chart containing the point  $\mathbf{o}_1$  which represents a closed orbit (cf. Lemma 2.2). Take the local equation  $\epsilon' = a_2$  in (12) to be the generator of the exceptional divisor ideal. Thus the blowup map is given by the relations,

$$\begin{aligned} a_3 &= b_1 + c_1\epsilon', \\ a_4 &= a_1b_1 + c_2\epsilon', \\ b_2 &= c_3\epsilon', \end{aligned} \tag{13}$$

where  $a_1, a_2, b_1, b_3, c_1, c_2, c_3$  are local coordinates.

We compute next the saturation of the image sheaf of the multiplication map  $\mu$  pulled back to  $\mathbb{X}'$ , that we will denote by  $\mathcal{E}_3^{10}$ . Let  $\mathcal{M}$  be the submodule span by rows of the matrix  $M$  obtained from (11) by pullback via (13). The tenth row of  $M'$  ( $M'$  denote the pullback of  $M$  via (13)) is of the form  $a_2R_0$  (i.e., all entries in the row are divisible by  $a_2$ ), where  $R_0$  is a row matrix that has an entry equal to 1. Hence, the saturation  $\mathcal{E}_3^{10}$  is a locally split subsheaf of  $S_3\mathcal{G}$  of rank 10.  $\square$

**Lemma 2.9.** *With notation as in Proposition 2.8. Let  $\mathbb{E}'$  denotes the exceptional divisor in the blowup of  $\mathbb{X}$  along  $\mathbb{Y}$ . Then we have*

1. *For  $\mathbf{x}' \in \mathbb{X}' - \mathbb{E}'$  lying over  $\mathbf{x} = (\ell, h; \mathbf{o}; \bar{q}) \in \mathbb{X} - \mathbb{Y}$ , the fiber of  $\mathcal{E}_3^{10}$  is*

$$(\mathcal{E}_3^{10})_{\mathbf{x}'} = h \cdot x_1 \cdot \mathcal{G} + h \cdot x_2 \cdot \mathcal{G} + q \cdot \mathcal{G} \text{ with } \ell = \langle x_1, x_2 \rangle \text{ and } \bar{q} \in S_2\mathcal{G}^\ell / (h \cdot \ell).$$

2. *Given  $\mathbf{x}' \in \mathbb{E}'$  lying over  $(\ell, h; \mathbf{o}; \overline{h h_p}) \in \mathbb{Y}$ . Let  $\mathbf{p}, \mathbf{o} \in \ell$  be the points defined by the ideals  $\langle x_1, x_2, h_p \rangle$  and  $\langle x_1, x_2, h_o \rangle$ , respectively. Then we have*

$$(\mathcal{E}_3^{10})_{\mathbf{x}'} = h(S_2\mathcal{G})_{\mathbf{p}} + \langle f_3 \rangle$$

*for a unique  $f_3 = h_2 \cdot f_2$  where  $f_2 \in \mathbb{P}((S_2\mathcal{G})_{\mathbf{o}, \mathbf{p}} / h \cdot \mathcal{G})$ . Here  $(S_2\mathcal{G})_{\mathbf{p}}$ ,  $(S_2\mathcal{G})_{\mathbf{o}, \mathbf{p}}$  denotes the space of quadratic forms in the ideal  $\langle x_1, x_2, h_p \rangle$ ,  $\langle x_1, x_2, h_o h_p \rangle$ , respectively, and the factor  $h_2$  in  $f_3$  stands for any equation*

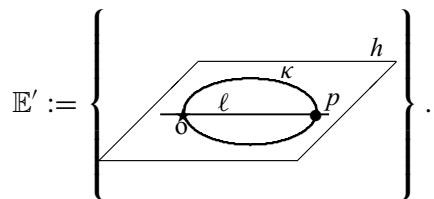
$$h_2 \in \mathbb{P}\left(\frac{\mathcal{L}_\ell}{h}\right) = \mathbb{P}\left(\frac{\langle x_1, x_2 \rangle}{h}\right).$$

**Proof.** Restricting to the exceptional divisor,  $\mathbb{E}'$ , we have that the new cubic (i.e. the last nonzero row of the local matrix representation of  $M'$ ) is given by  $x_2 \cdot f_2$ , where

$$f_2 = -x_2^2 - c_1x_2x_3 - c_2x_2x_4 + c_3x_3^2 + c_3(a_1 + b_3)x_3x_4 + a_1c_3b_3x_4^2. \quad (14)$$

is a conic in the plane  $h$  passing through the points  $o = [0 : 0 : -b_3 : 1]$  and  $p = [0 : 0 : -a_1 : 1]$ .  $\square$

**Remark 2.10.** We have that the cubic scroll determined by a point in  $\mathbb{E}'$  correspond to the union of a line  $\ell \subset h$  with a conic  $\kappa \subset h$  through the points  $o$  and  $p$  ( $\kappa$  is determined by  $f_2$ , see (14)), here  $p$  appears as an embedded point.



### 3 Cycle classes

Now we define the divisors that will appear when we will study the condition of a cone to be incident to a given fixed line in  $\mathbb{P}^n$ . A good exposition about cones and how the condition of a line to be incident to a cone is equivalent to the condition to the given line to be incident to the base of the cone is given by *Levcovitz-Vainsencher-Xavier* in Section 2 of [2]. So, since our base curve has two components (a line and a conic), we will determine the cycle class associated to the following two divisors.

Fix a line  $\ell_0$  in  $\mathbb{P}^n$  and remember that a point  $x \in \mathbb{X}$  correspond to  $(V, (\ell, h; o; \bar{q}))$  where  $o \in \ell \cap h$ ,  $(\ell, h) \in \mathbb{G}(2, \mathcal{G}) \times \mathbb{P}(\mathcal{G})$  and  $\bar{q} \in \mathbb{P}(S_2 \mathcal{G}^\ell / h \cdot \ell)$  with  $\mathcal{G} = \mathcal{T}_V$ . Let

$$\mathbb{D}_{\ell_0, \ell} = \left\{ (V, (\ell, h; o; \bar{q})) \in \mathbb{X} \mid \ell \cap \ell_0 \neq \emptyset \right\},$$

$$\mathbb{D}_{\ell_0, \kappa} = \left\{ (V, (\ell, h; o; \bar{\kappa})) \in \mathbb{X} \mid \kappa \cap \ell_0 \neq \emptyset \right\}.$$

**Proposition 3.1.** *The cycle classes of divisors defined above can be expressed as follows in terms of Chern classes of the natural bundles in Subsection 2.1:*

1.  $[\mathbb{D}_{\ell_0, \ell}] = -c_1(\mathcal{L}_{\mathcal{T}}).$
2.  $[\mathbb{D}_{\ell_0, \kappa}] = c_1(\mathcal{O}_{\mathcal{T}}(1)) + 2c_1(\mathcal{O}_{\mathcal{T}}(1)).$

**Proof.** Let  $U \subset \mathbb{G}[v, n]$  be the open set defined by the condition  $\ell_0 \cap \mathbf{V} = \emptyset$  (i.e. the vertices in  $U$  are not incident to the line  $\ell_0$ ).

Let us now consider the pull-back to  $\mathbb{X}$  of the tautological sequence in (3) (denoted in the same way)

$$0 \longrightarrow \mathcal{O}_{\mathcal{T}}(-1) \longrightarrow \mathcal{T} \longrightarrow \mathcal{H} \longrightarrow 0.$$

Set  $L_0 \subset \mathcal{T}$  be the subbundle (over  $U$ ) whose fiber is given by the equations of the projection of the fixed line  $\ell_0 \subset \mathbb{P}^n$ . And note that the quotient bundle

$$\overline{L_0} = \frac{\mathcal{T}}{L_0} \cong \mathbb{C}^2$$

is trivial. Then consider the composition of the map  $\mathcal{O}_{\mathcal{T}}(-1) \longrightarrow \mathcal{T} \longrightarrow \overline{L_0}$  over the transversal locus given by the condition that the hyperplane does not contain the line. The quotient map

$$\mathcal{T} \longrightarrow \overline{\mathcal{H}} = \frac{\mathcal{T}}{\mathcal{O}_{\mathcal{T}}(-1) \oplus L_0}$$

can be thought of as evaluation of linear forms on  $\mathcal{T}$  at the point of intersection of the projection of the line  $\ell_0$  with the varying transversal hyperplane. Its induced a map of quadratic forms  $S_2\mathcal{H} \longrightarrow S_2\overline{\mathcal{H}}$ .

According to notations in Section 2.1 we have the following diagram of fiber bundles over  $\mathbb{X}$ .

$$\begin{array}{ccccccc} & & 0 & & & & \\ & & \downarrow & & & & \\ & \mathcal{L} \otimes \mathcal{O}_{\mathcal{T}}(-1) & = & \mathcal{L} \otimes \mathcal{O}_{\mathcal{T}}(-1) & & & \\ & \downarrow & & \downarrow & & & \\ 0 \longrightarrow & S_2\mathcal{T}^\perp & \longrightarrow & S_2\mathcal{T} & \longrightarrow & S_2\overline{\mathcal{T}} \longrightarrow 0 \\ & \downarrow & & \downarrow & & & \\ & \mathcal{E} & \longrightarrow & \frac{S_2\mathcal{T}}{\mathcal{L} \otimes \mathcal{O}_{\mathcal{T}}(-1)} & \longrightarrow & \frac{S_2\mathcal{T}}{\mathcal{L} \otimes \mathcal{T}} = S_2\left(\frac{\mathcal{T}}{\mathcal{O}_{\mathcal{T}}(-1)}\right) = S_2\mathcal{H} \\ & \downarrow & & & & & \\ & 0 & & & & & \end{array}$$

Composing the inclusion map  $\mathcal{O}_{\mathcal{E}}(-1) \hookrightarrow \mathcal{E}$  with the sequence of maps in the latter line above and the map  $S_2\mathcal{H} \longrightarrow S_2\overline{\mathcal{H}}$ . The latter vanished precisely over  $\mathbb{D}_{\ell_0, \kappa}$ . In fact, we have that  $[\mathbb{D}_{\ell_0, \kappa}] = c_1(\mathcal{O}_{\mathcal{E}}(1) \otimes S_2\overline{\mathcal{H}}) = c_1(\mathcal{O}_{\mathcal{E}}(1)) + 2c_1(\mathcal{O}_{\mathcal{T}}(1))$ .

On the other hand composing the inclusion  $\mathcal{L}_{\mathcal{T}} \longrightarrow \mathcal{T}$  with the quotient map  $\mathcal{T} \longrightarrow \overline{L_0}$ , we verified that the second exterior power vanished over  $\mathbb{D}_{\ell_0, \ell}$ . Thus,  $[\mathbb{D}_{\ell_0, \ell}] = c_1(\bigwedge^2 \mathcal{L}^\vee) = -c_1(\mathcal{L})$  (here we use that  $\overline{L_0}$  is a trivial bundle).  $\square$

Thus the cycle  $\mathbb{D} := \mathbb{D}_{\ell_0, \ell} + \mathbb{D}_{\ell_0, \kappa}$  parametrized cones incident to a line.

**Cycles in  $\mathbb{X}'$ .** Note that the cycle  $\mathbb{D}_{\ell_0, \ell}$  does not meet the blow-up center  $\mathbb{Y}$  (cf. (8)), then its strict transform is isomorphic to itself. On the other hand the cycle  $\mathbb{D}_{\ell_0, \kappa}$  meet the blow-up center  $\mathbb{Y}$ , and local computations along suitable coordinate chart show that the total transform of  $\mathbb{D}_{\ell_0, \kappa}$  is equal to  $\mathbb{D}'_{\ell_0, \kappa} + \mathbb{E}$ . Thus

$$[\mathbb{D}'] = [\mathbb{D}_{\ell_0, \ell}] + [\mathbb{D}_{\ell_0, \kappa}] - \mathbb{E}$$

and we wish to compute the degree

$$\int_{\mathbb{X}'} ([\mathbb{D}] - [\mathbb{E}])^d \quad (15)$$

where  $d = \dim \mathbb{X}' = (v+1)(n-v) + 5(n-v-1) - 4$ .

#### 4 Applying Bott's formula

We made the computation of the degree in (15) by means of Bott's localization formula in equivariant cohomology (cf. [1] and p. 35 in [4]).

**Bott's formula.** Given a variety  $\mathbb{V}$  endowed with a  $\mathbb{C}^*$ -action, suppose that the set of fixed points is finite. Let  $p$  be a polynomial in the Chern classes of equivariant vector bundles over  $\mathbb{V}$  of weighted degree equal to the dimension of  $\mathbb{V}$ . Then we have

$$\int_{\mathbb{V}} p = \sum_x \frac{p_x^{eq}}{c_x^{eq}(T\mathbb{V})}. \quad (16)$$

The sum ranges over the set of fixed points; the numerator  $p_x^{eq}$  denotes the equivariant class at the fixed point  $x$ , and likewise for the equivariant top Chern class of the tangent bundle,  $T\mathbb{V}$ , occurring in the denominator. These equivariant classes translate into appropriate symmetric functions on the weights of the fibers of the relevant bundles at each fixed point as explained below.

**$\mathbb{C}^*$ -action.** Let  $\mathcal{F}$  be the space of linear forms in  $\mathbb{P}^n$  with homogeneous coordinates  $x_1, \dots, x_{n+1}$ . We assume the  $\mathbb{C}^*$ -action is given by sufficiently general weights  $w_1, \dots, w_{n+1}$ . This means that the action can be described by

$$t \cdot x_i = t^{w_i} x_i, \quad 1 \leq i \leq n+1, \quad t \in \mathbb{C}^*.$$

Doing a generic choice of the weights  $w_1, \dots, w_{n+1}$  ( $w_i \neq w_j$  for  $i \neq j$ ), we have induced an action over  $G[v, n]$  obtaining only isolated fixed points. Also this action induces an action on  $\mathbb{X}$  and  $\mathbb{X}'$ , respectively, having only isolated fixed points.

From now on, for the sake of concreteness and simplicity, we shall assume that  $n = 4$  and  $v = 0$  (so  $d = 15$ ).

**Fixed points.** The grassmannian of vertices  $\mathbb{G}[v, n] = \mathbb{P}^4$  has the 5 fixed points  $\{[e_i]\}_{i=1}^5$  ( $\{e_i\}_{i=1}^5$  denotes the canonical base of  $\mathbb{C}^5$ ). Fix the vertex  $\mathbf{V} = [0 : 0 : \cdots 0 : 1]$  (the forthcoming computations also apply to the other vertices doing and appropriate permutation). Thus  $\mathcal{G} = \mathcal{T}_{\mathbf{V}} = [x_1, x_2, x_3, x_4]$ . Therefore have the six fixed points  $[x_i, x_j]$  for  $1 \leq i < j \leq 4$  in  $\mathbb{G}(2, \mathcal{G})$  and the 4 fixed points  $[x_i]$  for  $1 \leq i \leq 4$  in  $\mathbb{P}(\mathcal{G})$ , so at all we have  $6 \cdot 4 = 24$  fixed points in  $\mathbb{G}(2, \mathcal{G}) \times \mathbb{P}(\mathcal{G})$ . By inspection we see that exactly 12 of those fixed points belong to  $\Sigma$  (cf. (5)), namely  $([x_i, x_j], [x_k])$  where for each choice of  $(i, j)$  with  $1 \leq i < j \leq 4$  we take  $k \in \{i, j\}$ . Knowing that the blow-up center  $\mathbb{Y}$  (cf. (8)) lies over  $\Sigma$  we conclude that the other 12 fixed points  $([x_i, x_j], [x_k])$  (where for each choice of  $(i, j)$  with  $1 \leq i < j \leq 4$  we take  $k \notin \{i, j\}$ ) go isomorphically to  $12 \cdot 5 = 60$  fixed points in  $\mathbb{X}'$  since the fiber of  $\mathcal{T}$  over each of those fixed points is formed by 5 fixed points (cf. (4)). Therefore, they contribute immediately to Bott's formula.

Now for the sake of concreteness, we pick the fixed point  $(\ell, h) \in \Sigma$  where  $\ell = [x_1, x_2]$  and  $h = [x_1]$ . Note that the fiber of  $\mathbb{E}$  over  $(\ell, h)$  is equal to  $[x_1, x_2, x_3, x_4]/[x_1, x_2] = [\overline{x_3}, \overline{x_4}]$  (cf. Remarks 2.1). Then over each fixed point in  $\Sigma$  we have exactly 2 fixed points in  $\mathbb{E}$ . Thus in  $\mathbb{E}$  we have the 24 fixed points  $\underbrace{([x_i, x_j], [x_k]; [\overline{x_s}])}_{\text{in } \Sigma}$  where  $s \in \{1, \dots, 4\} \setminus \{i, j\}$ .

According to (4) the fiber of  $\mathcal{T}$  over  $(\ell, h) \in \Sigma$  is equal to

$$[\overline{x_1 x_3}, \overline{x_1 x_4}, \overline{x_2^2}, \overline{x_2 x_3}, \overline{x_2 x_4}].$$

And we have the same fiber for the two fixed points in  $\mathbb{E}$  that lies over the fixed point  $(\ell, h) \in \Sigma$ . Set  $\mathbf{o} = [\overline{x_3}]$  then again by inspection on the fiber of  $\mathcal{T}$  over  $(\ell, h; \mathbf{o}) \in \mathbb{E}$  we verified that the 3 fixed points  $((\ell, h; \mathbf{o}; \overline{\kappa}))$  with  $\kappa \in \{x_2^2, x_2 x_3, x_2 x_4\}$  are not in  $\mathbb{Y}$  and the 2 fixed points  $((\ell, h; \mathbf{o}; \overline{\kappa}))$  with  $\kappa \in \{x_1 x_3, x_1 x_4\}$  belong to  $\mathbb{Y}$ . Therefore, we have (doing permutations) exactly  $12 \cdot 2 \cdot 3 = 72$  fixed points in  $\mathbb{X} \setminus \mathbb{Y}$  (lying over  $\Sigma$ ) going isomorphically to fixed points in  $\mathbb{X}'$  that contribute immediately to Bott's formula and  $12 \cdot 2 \cdot 2 = 48$  fixed points in  $\mathbb{Y}$  (lying over  $\Sigma$ ). Therefore we find a total of

$$\underbrace{12 \cdot 5}_{\text{Mapping off } \Sigma} + \underbrace{12 \cdot 2 \cdot 3}_{\text{Off } \mathbb{Y} \text{ mapping in } \Sigma} + \underbrace{12 \cdot 2 \cdot 2}_{\text{In } \mathbb{Y} \text{ mapping in } \Sigma} = 180 \text{ fixed points in } \mathbb{X}$$

(for a fixed vertex), then varying the vertex we have a total of  $180 \cdot 5 = 900$  fixed points in  $\mathbb{X}$ .

Fix the vertex  $\mathbf{V} = [0 : 0 : \cdots 0 : 1]$  again. Then we have  $132 (= 12 \cdot 2 \cdot 3 + 12 \cdot 5)$  fixed points going isomorphically to 132 fixed points in  $\mathbb{X}' \setminus \mathbb{E}'$

and  $192 = 12 \cdot 2 \cdot 2 \cdot 4$  fixed points in  $\mathbb{E}'$  since the fiber of  $\mathbb{E}'$  over the fixed points  $(\ell, h; o; q)$  with  $q = \overline{x_1 x_3}$  is given by  $[x_2^3, x_2 x_3, x_2^2 x_4, x_2 x_3^2]$  according to Lemma 2.9. Therefore we find a total of 324 fixed points in  $\mathbb{X}'$  (for a fixed vertex), then varying the vertex we have a total of  $324 \cdot 5 = 1620$  fixed points in  $\mathbb{X}'$ .

**Weights.** Now we describe the weights of the induced  $\mathbb{C}^*$ -representation in the fibers of the relevant bundles at each fixed point. Next by convenience we write the weights decomposition in the form

$$\mathcal{F} = t^{w_1} + \dots + t^{w_5}.$$

Each summand of  $t^{w_i}$  above stands for the eigensubspace of  $\mathcal{F}$  where the action is given by the induced character. In the final step we have to replace it  $w_i$  by suitable functions of the original weights, say  $x_1, \dots, x_5$  of the  $\mathbb{C}^*$ -action on  $\mathbb{P}^4$ , keeping track of the relevant induced actions.

*Weights for fixed points in  $\mathbb{X}'$  mapping off  $\Sigma$ .* Consider the fixed point  $\mathbf{x}' = (\mathbf{V}, (\ell, h; \kappa))$  where  $\mathbf{V} = [0 : \dots : 0 : 1]$ ,  $\ell = [x_1, x_2]$ ,  $h = [x_3]$  and  $\kappa = \overline{x_1^2} \in \mathcal{G}^\ell / (\ell \cdot h) = \mathcal{E}_{(\ell, h)}$  (cf. (4)) ( $G = \mathcal{T}_{\mathbf{V}} = [x_1, x_2, x_3, x_4]$ ). Thus to determine the contributions to Bott's formula in this case, since the exceptional divisor  $\mathbb{E}'$  give no contribution here, we have only to determine for the numerator of the right-hand side of (16), specifically, the following equivariant Chern classes

$$c_1^T((\mathcal{L}_{\mathcal{T}})_{(\mathbf{V}; \ell)}), \quad c_1^T(\mathcal{O}_{\mathcal{E}(1)_{\mathbf{x}}}) \quad (\mathbf{x} := \mathbf{x}' \text{ since } \mathbb{X}' \setminus \mathbb{E}' \cong \mathbb{X} \setminus \mathbb{Y})$$

and  $c_1^T(\mathcal{O}_{\mathcal{T}(1)_h})$ . In fact, we have

$$(\mathcal{L}_{\mathcal{T}})_{(\mathbf{V}; \ell)} = t^{w_1} + t^{w_2} \quad \text{and} \quad c_1^T((\mathcal{L}_{\mathcal{T}})_{(\mathbf{V}; \ell)}) = w_1 + w_2,$$

$$\mathcal{O}_{\mathcal{E}(1)_{\mathbf{x}}} = t^{-2w_1} \quad \text{and} \quad c_1^T(\mathcal{O}_{\mathcal{E}(1)_{\mathbf{x}}}) = -2w_1,$$

$$\mathcal{O}_{\mathcal{T}(1)_h} = t^{-w_3} \quad \text{and} \quad c_1^T(\mathcal{O}_{\mathcal{T}(1)_h}) = -w_3.$$

For the denominator of the right-hand side of (16), we have that

$$T_{\mathbb{X}'_{\mathbf{x}'}} = T_{\mathbf{x}} \mathbb{X} = T_{(\mathbf{V}, (\ell, h))} \mathbb{B} + T_{\mathbf{x}} \mathbb{P}(\mathcal{E}_{(\mathbf{V}, (\ell, h))}) + T_{\mathbf{V}} \mathbb{P}^4$$

and the last term above is equal to

$$T_{(\mathbf{V}, \ell)} \mathbb{G}(2, \mathcal{T}) + T_{(\mathbf{V}, h)} \mathbb{P}(\mathcal{T}) + T_{\mathbf{x}} \mathbb{P}(\mathcal{E}_{(\mathbf{V}, (\ell, h))}) + T_{\mathbf{V}} \mathbb{P}^4.$$

Thus

$$\begin{aligned}
 T_x \mathbb{X} &= (\mathcal{L}_{\mathcal{T}})_{(\mathbf{V}, \ell)}^{\vee} \otimes \overline{\mathcal{T}}_{(\mathbf{V}, \ell)} + \mathcal{O}_{\mathcal{T}}(-1)_{(\mathbf{V}, h)}^{\vee} \otimes \mathcal{H}_{(\mathbf{V}, h)} \\
 &\quad + \mathcal{O}_{\mathcal{E}}(-1)_x^{\vee} \otimes (\mathcal{E}/\mathcal{O}_{\mathcal{E}}(-1))_x + \mathcal{O}_{\mathbb{P}^4}(-1)_{\mathbf{V}}^{\vee} \otimes (\mathbb{C}^5/\mathcal{O}_{\mathbb{P}^4}(-1))_{\mathbf{V}} \\
 &= (t^{-w_1} + t^{-w_2})(t^{w_3} + t^{w_4}) + t^{-w_3}(t^{w_1} + t^{w_2} + t^{w_4}) + t^{-2w_1}(t^{(w_1+w_2)} \\
 &\quad + t^{(w_1+w_4)} + t^{2w_2} + t^{(w_2+w_4)}) + (t^{-w_1} + t^{-w_2} + t^{-w_3} + t^{-w_4})t^{w_5} \\
 &= t^{(w_3-w_1)} + t^{(w_4-w_1)} + t^{(w_3-w_2)} + t^{(w_4-w_2)} + t^{(w_1-w_3)} + t^{(w_2-w_3)} \\
 &\quad + t^{(w_4-w_3)} + t^{(w_2-w_1)} + t^{(w_4-w_1)} + t^{2(w_2-w_1)} + t^{(w_2+w_4-2w_1)} \\
 &\quad + t^{(w_5-w_1)} + t^{(w_5-w_2)} + t^{(w_5-w_3)} + t^{(w_5-w_4)}.
 \end{aligned}$$

Therefore

$$\begin{aligned}
 c_{15}^T(T\mathbb{X}'_{x'}) &= 2(w_3 - w_1)^2(w_4 - w_1)^2(w_3 - w_2)^2(w_4 - w_2)(w_4 - w_3)(w_2 - w_1)^2 \\
 &\quad \times (w_2 + w_4 - 2w_1)(w_5 - w_1)(w_5 - w_2)(w_5 - w_3)(w_5 - w_4).
 \end{aligned}$$

*Weights for fixed points in  $\mathbb{X}' \setminus \mathbb{E}'$  mapping in  $\Sigma$ .* Consider the fixed point  $x' = (\mathbf{V}, (\ell, h; \mathbf{o}; \kappa))$  where  $\mathbf{V} = [0 : \dots : 0 : 1]$ ,  $\ell = [x_1, x_2]$ ,  $h = [x_1]$ ,  $\mathbf{o} = [\overline{x_3}] \in \mathbb{P}(\mathcal{G}/\ell)$  and  $\kappa = \overline{x_2^2} \in \mathcal{G}^\ell/(\ell \cdot h) = \mathcal{E}_{(\ell, h)}$  (cf. (4)) ( $\mathcal{G} = \mathcal{T}_{\mathbf{V}} = [x_1, x_2, x_3, x_4]$ ). Thus to determine the contributions to Bott's formula in this case, since the exceptional divisor  $\mathbb{E}'$  give no contribution here, we have only to determine for the numerator of the right-hand side of (16), specifically, the following equivariant Chern classes  $c_1^T((\mathcal{L}_{\mathcal{T}})_{(\mathbf{V}, \ell)})$ ,  $c_1^T(\mathcal{O}_{\mathcal{E}}(1)_x)$  ( $x := x'$  since  $\mathbb{X}' \setminus \mathbb{E}' \cong \mathbb{X} \setminus \mathbb{Y}$ ) and  $c_1^T(\mathcal{O}_{\mathcal{T}}(1)_h)$ . In fact, we have

$$\begin{aligned}
 (\mathcal{L}_{\mathcal{T}})_{(\mathbf{V}, \ell)} &= t^{w_1} + t^{w_2} \quad \text{and} \quad c_1^T((\mathcal{L}_{\mathcal{T}})_{(\mathbf{V}, \ell)}) = w_1 + w_2, \\
 \mathcal{O}_{\mathcal{E}}(1)_x &= t^{-2w_2} \quad \text{and} \quad c_1^T(\mathcal{O}_{\mathcal{E}}(1)_x) = -2w_2, \\
 \mathcal{O}_{\mathcal{T}}(1)_h &= t^{-w_1} \quad \text{and} \quad c_1^T(\mathcal{O}_{\mathcal{T}}(1)_h) = -w_1.
 \end{aligned}$$

For the denominator of the right-hand side of (16), we have that

$$T\mathbb{X}'_{x'} = T_x \mathbb{X} = T_{(\mathbf{V}, (\ell, h; \mathbf{o}))} \mathbb{B} + T_x \mathbb{P}(\mathcal{E}_{(\mathbf{V}, (\ell, h; \mathbf{o}))}) + T_{\mathbf{V}} \mathbb{P}^4. \quad (17)$$

In fact

$$\begin{aligned}
 T_{(\mathbf{V}, (\ell, h; \mathbf{o}))} \mathbb{B} &= T_{(\mathbf{V}, (\ell, h))} \Sigma + T_{(\mathbf{V}, (\ell, h; \mathbf{o}))} \mathbb{E}_{(\mathbf{V}, (\ell, h))} + \mathcal{O}(\mathbb{E})_{(\mathbf{V}, (\ell, h; \mathbf{o}))} \\
 &= \underbrace{T_{(\mathbf{V}, \ell)} \mathbb{G}(2, \mathcal{T}) + T_{(\mathbf{V}, (\ell, h))} \mathbb{P}(\mathcal{G}^\ell)}_{= T_{(\mathbf{V}, (\ell, h))} \Sigma} + \underbrace{T_{(\mathbf{V}, (\ell, h; \mathbf{o}))} \mathbb{P}(\mathcal{G}/\ell)}_{= T_{(\mathbf{V}, (\ell, h; \mathbf{o}))} \mathbb{E}_{(\mathbf{V}, (\ell, h))}} + \underbrace{\mathcal{O}_{\mathbb{E}}(-1)_{(\mathbf{V}, (\ell, h; \mathbf{o}))}}_{= \mathcal{O}(\mathbb{E})_{(\mathbf{V}, (\ell, h; \mathbf{o}))}} \\
 &= \underbrace{(t^{-w_1} + t^{-w_2})(t^{w_3} + t^{w_4})}_{T_{(\mathbf{V}, \ell)} \mathbb{G}(2, \mathcal{T})} + \underbrace{t^{-w_1} \cdot t^{w_2}}_{T_{(\mathbf{V}, (\ell, h))} \mathbb{P}(\mathcal{G}^\ell)} + \underbrace{t^{-w_3} \cdot t^{w_4}}_{T_{(\mathbf{V}, (\ell, h; \mathbf{o}))} \mathbb{P}(\mathcal{G}/\ell)} + \underbrace{t^{-w_1} \cdot t^{w_3}}_{\mathcal{O}_{\mathbb{E}}(-1)_{(\mathbf{V}, (\ell, h; \mathbf{o}))}}.
 \end{aligned} \quad (18)$$



Note that the fiber of  $\mathcal{O}(\mathbb{E})_{\mathbb{E}}$  over  $\mathbf{b} := (\mathbf{V}, (\ell, h; \mathbf{o}))$  is the subspace of  $(\mathcal{N}_{\Sigma/\mathbb{G}(2, \mathcal{T}) \times \mathbb{P}(\mathcal{T})})_{(\mathbf{V}, (\ell, h))}$  that correspond (tautologically) to  $\mathbf{b}$ . In order to find it, we compare the following two descriptions of the exceptional fiber

$$\mathbb{E}_{(\mathbf{V}, (\ell, h))} = \mathbb{P}(\mathcal{G}/\ell) = \mathbb{P}([\bar{x}_3, \bar{x}_4]) \quad \text{and} \quad \mathbb{P}((\mathcal{N}_{\Sigma/\mathbb{G}(2, \mathcal{T}) \times \mathbb{P}(\mathcal{T})})_{(\mathbf{V}, (\ell, h))}).$$

Here we have

$$\begin{aligned} (\mathcal{N}_{\Sigma/\mathbb{G}(2, \mathcal{T}) \times \mathbb{P}(\mathcal{T})})_{(\mathbf{V}, (\ell, h))} &= T_{(\mathbf{V}, (\ell, h))} \mathbb{G}(2, \mathcal{T}) \times \mathbb{P}(\mathcal{T}) - T_{(\mathbf{V}, (\ell, h))} \Sigma \\ &= t^{-w_1} (t^{w_3} + t^{w_4}), \end{aligned}$$

and we check that the correspondence is given by  $[\bar{x}_3] \leftrightarrow t^{w_3-w_1}$  and  $[\bar{x}_4] \leftrightarrow t^{w_4-w_1}$ .

By analogous computation we obtain that:

$$\begin{aligned} T_{\mathbf{V}} \mathbb{P}(\mathcal{E}_{(\mathbf{V}, (\ell, h; \mathbf{o}))}) &= t^{-2w_2} (t^{(w_1+w_3)} + t^{(w_1+w_4)} + t^{(w_2+w_3)} + t^{(w_2+w_4)}), \\ T_{\mathbf{V}} \mathbb{P}^4 &= (t^{-w_1} + t^{-w_2} + t^{-w_3} + t^{-w_4}) t^{w_5}. \end{aligned} \quad (19)$$

Therefore

$$\begin{aligned} c_{15}^T(T\mathbb{X}'_{\mathbf{x}'}) &= (w_3 - w_1)^2 (w_4 - w_1) (w_3 - w_2)^2 (w_4 - w_2)^2 (w_4 - w_3) (w_2 - w_1) \\ &\quad \times (w_1 + w_4 - 2w_2) (w_5 - w_1) (w_5 - w_2) (w_5 - w_3) (w_5 - w_4) \\ &\quad \times (w_1 + w_3 - 2w_2). \end{aligned}$$

*Weights for fixed points in  $\mathbb{E}'$ .* Consider the fixed point  $\mathbf{x}' = (\mathbf{V}, (\ell, h; \mathbf{o}; \kappa; \mathbf{c}))$  where  $\mathbf{V} = [0 : \dots : 0 : 1]$ ,  $\ell = [x_1, x_2]$ ,  $h = [x_1]$ ,  $\mathbf{o} = [\bar{x}_3] \in \mathbb{P}(\mathcal{G}/\ell)$ ,  $\kappa = \overline{x_1 x_3} \in \mathcal{G}^\ell/(\ell \cdot h) = \mathcal{E}_{(\ell, h)}$  (cf. (4)) and  $\mathbf{c} = \bar{x}_2^3 \in (x_2 \cdot (S_2 \mathcal{G})_{\mathbf{o}, \mathbf{p}})/(x_1 \cdot S_2 \mathcal{G})$  (cf. Lemma 2.9) ( $\mathcal{G} = \mathcal{T}_{\mathbf{V}} = [x_1, x_2, x_3, x_4]$ ). Thus to determine the contributions to Bott's formula in this case, since the exceptional divisor  $\mathbb{E}'$  give contribution here, we have to determine for the numerator of the right-hand side of (16), the following equivariant Chern classes  $c_1^T((\mathcal{L}_{\mathcal{T}})_{(\mathbf{V}; \ell)})$ ,  $c_1^T(\mathcal{O}_{\mathcal{E}}(1)_{\mathbf{x}})$  with  $\mathbf{x} = (\mathbf{V}, (\ell, h; \mathbf{o}; \kappa))$ ,  $c_1^T(\mathcal{O}_{\mathcal{T}}(1)_h)$  and  $c_1^T(\mathcal{O}(\mathbb{E}')_{\mathbf{x}'})$ . In fact, we have

$$\begin{aligned} (\mathcal{L}_{\mathcal{T}})_{(\mathbf{V}; \ell)} &= t^{w_1} + t^{w_2} \quad \text{and} \quad c_1^T((\mathcal{L}_{\mathcal{T}})_{(\mathbf{V}; \ell)}) = w_1 + w_2, \\ \mathcal{O}_{\mathcal{E}}(1)_{\mathbf{x}} &= t^{-w_1-w_3} \quad \text{and} \quad c_1^T(\mathcal{O}_{\mathcal{E}}(1)_{\mathbf{x}}) = -w_1 - w_3, \\ \mathcal{O}_{\mathcal{T}}(1)_h &= t^{-w_1} \quad \text{and} \quad c_1^T(\mathcal{O}_{\mathcal{T}}(1)_h) = -w_1. \end{aligned}$$

Note that the fiber  $\mathcal{O}(\mathbb{E}')_{\mathbf{x}'}$  is the subspace of  $(\mathcal{N}_{\mathbb{Y}/\mathbb{X}})_{\mathbf{x}'}$  that correspond (tautologically) to  $\mathbf{x}'$ . In order to find it, we compare the following two descriptions

of the exceptional fiber

$$\begin{aligned}\mathbb{E}'_x &= \mathbb{P}((x_2 \cdot (S_2\mathcal{G})_{\mathfrak{o},\mathfrak{p}})/(x_1 \cdot S_2\mathcal{G})) \\ &= \mathbb{P}(\overline{[x_2^3, x_2^2x_3, x_2^2x_4, x_2x_3^2]}) \text{ and } \mathbb{P}(\mathcal{N}_{\mathbb{Y}/\mathbb{X}}(\mathbf{V}, (\ell, h; \mathfrak{o}; \kappa))).\end{aligned}$$

Here we have that  $(\mathcal{N}_{\mathbb{Y}/\mathbb{X}})_x = T_x\mathbb{X} - T_x\mathbb{Y}$  with  $x = (\mathbf{V}, (\ell, h; \mathfrak{o}; \kappa))$ . According to (17), (18), (19) and knowing that

$$T_x\mathbb{P}(\mathcal{E}_{(\mathbf{V}, (\ell, h; \mathfrak{o}))}) = t^{(-w_1-w_3)}(t^{(w_1+w_4)} + t^{2w_2} + t^{(w_2+w_3)} + t^{(w_2+w_4)}).$$

We conclude that

$$\begin{aligned}T_x\mathbb{X} &= (t^{-w_1} + t^{-w_2})(t^{w_3} + t^{w_4}) + t^{-w_1} \cdot t^{w_2} + t^{-w_3} \cdot t^{w_4} + t^{-w_1} \cdot t^{w_3} \\ &\quad + t^{(-w_1-w_3)}(t^{(w_1+w_4)} + t^{2w_2} + t^{(w_2+w_3)} + t^{(w_2+w_4)}) \\ &\quad + (t^{-w_1} + t^{-w_2} + t^{-w_3} + t^{-w_4})t^{w_5}.\end{aligned}$$

On the other hand from the description of  $\mathbb{Y}$  in (8) we conclude that

$$\begin{aligned}T_x\mathbb{Y} &= T_{(\mathbf{V}, (\ell, h))}\Sigma + T_x\mathbb{E}_{(\mathbf{V}, (\ell, h))} + T_x\mathbb{P}(h \cdot (\mathcal{G}/\ell)) + T_{\mathbf{V}}\mathbb{P}^4 \\ &= (t^{-w_1} + t^{-w_2})(t^{w_3} + t^{w_4}) + t^{-w_1} \cdot t^{w_2} + t^{-w_3} \cdot t^{w_4} \\ &\quad + t^{(-w_1-w_3)} \cdot t^{(w_1+w_4)} + (t^{-w_1} + t^{-w_2} + t^{-w_3} + t^{-w_4})t^{w_5}\end{aligned}$$

Therefore

$$(\mathcal{N}_{\mathbb{Y}/\mathbb{X}})_x = t^{-w_1} \cdot t^{w_3} + t^{(-w_1-w_3)}(t^{2w_2} + t^{(w_2+w_3)} + t^{(w_2+w_4)}).$$

In this case we check that the correspondence is given by

$$\overline{x_2^3} \leftrightarrow t^{(2w_2-w_1-w_3)}, \overline{x_2^2x_3} \leftrightarrow t^{w_2-w_1}, \overline{x_2^2x_4} \leftrightarrow t^{(w_2+w_4-w_1-w_3)}, \overline{x_2x_3^2} \leftrightarrow t^{w_3-w_1}.$$

Thus  $\mathcal{O}(\mathbb{E}'_{x'}) = t^{(2w_2-w_1-w_3)}$  and  $c_1^T(\mathcal{O}(\mathbb{E}'_{x'})) = 2w_2 - w_1 - w_3$ .

For the denominator of the right-hand side of (16), we have that

$$T\mathbb{X}'_{x'} = T_x\mathbb{Y} + T_{x'}\mathbb{E}'_x + \mathcal{O}(\mathbb{E}'_{x'}).$$

Using the computation of  $T_x\mathbb{Y}$ ,  $\mathcal{O}(\mathbb{E}'_{x'})$  and the description for  $\mathbb{E}'_x$  given before, we conclude that:

$$\begin{aligned}c_{15}^T(T\mathbb{X}'_{x'}) &= 2(w_3 - w_1)(w_4 - w_1)(w_3 - w_2)^3(w_4 - w_2)^2(w_4 - w_3)^2(w_2 - w_1) \\ &\quad \times (2w_2 - w_1 - w_3)(w_5 - w_1)(w_5 - w_2)(w_5 - w_3)(w_5 - w_4).\end{aligned}$$

In order to determine the degree in formula (15) above, we used a MAPLE script. We find the following table for the number of cones in  $\mathbb{P}^n$  for  $4 \leq n \leq 6$  with vertices of dimension  $v = 0, 1, 2$ , respectively. Making appropriate changes from Remarks 2.1, we can also consider vertices in  $\mathbb{P}^n$  of codimension 5. We put for short,  $rtwc$  = reducible twisted cubic (in  $\mathbb{P}^3$ ) and  $rcsc$  = reducible cubic surface (in  $\mathbb{P}^4$ ).

| base   | $v = \text{vertex dimension}$ | $n = \dim \mathbb{P}^n$ | $\int (\mathbb{D}')^d$ |
|--------|-------------------------------|-------------------------|------------------------|
| $rtwc$ | 0                             | 4                       | 352032844800           |
| $rtwc$ | 1                             | 5                       | 3702700362203136       |
| $rtwc$ | 2                             | 6                       | 33620454508602458112   |
| $rcsc$ | 0                             | 5                       | 27352396059770880      |
| $rcsc$ | 1                             | 6                       | 995290170879010406400  |

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## References

- [1] D. Edidin and W. Graham. *Localization in equivariant intersection theory and the Bott residue formula*. Am. J. Math., **120**(3) (1998), 619–636.
- [2] D. Levcovitz, I. Vainsencher and F. Xavier. *Enumeration of cones over cubic scrolls*. Israel J. Math., **161** (2007), 103–123.
- [3] J. Harris. *Algebraic Geometry*. Springer-Verlag New York (1992).
- [4] A. Meireles and I. Vainsencher. *Equivariant Intersection Theory and Bott's Residue Formula*. Notes for the Escola de Álgebra, UnB-2000, Mat. Contemporanea, SBM, **20**(1) (2001), 1–70.
- [5] J. Rojas and I. Vainsencher. *Canonical Curves in  $\mathbb{P}^3$* . Proc. London Math. Soc., **3**(85) (2002), 333–366.
- [6] T.A. Springer. *Linear Algebraic Groups*. Modern Birkhäuser Classics. 2<sup>nd</sup> ed. (1998).
- [7] I. Vainsencher and F. Xavier. *Numbers for reducible cubic scrolls*. Anais da Academia Brasileira de Ciências, **76**(4) (2004), 645–650.

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